

7(a)	When an electron absorbs the energy of a photon, it gains the necessary energy to be emitted from the metal surface. As it moves through the metal, it collides and interact with the lattice ions and other electrons and energy is lost. Since different electrons loses different amount of energies, the different electrons are then emitted with different kinetic energies.
7(b)	Using $hf = \phi + KE_{\max}$ $hf = \frac{hc}{\lambda_{\max}} + KE_{\max}$ $\lambda_{\max} = \frac{hc}{hf - KE_{\max}} = \frac{hc}{(7.95 - 3.48)(1.6 \times 10^{-19})}$ $= 2.78 \times 10^{-7} \text{ m}$
7(c)	With higher intensity, number of incident photons per second increases. Thus, more electrons are emitted per second and therefore the metal plate will take a shorter time to be discharged.
7(d)	Infrared radiation has shorter wavelength than ultraviolet(UV) radiation. As such, its frequency is lower than that of UV and thus is lower than the threshold frequency. As such, no electrons are emitted and the metal plate will not be discharged.
8(a)	The half life of 87.7 years is long enough to sustain power generation for the